

PARTHA GHOSH

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WORK EXPERIENCE

REVE Systems 📅 Jan 2024 – Present
Machine Learning Engineer

- Fine-tuned **Wav2vec 2.0** and **Whisper** model for **Bengali STT (কথা)**, also utilizing **PEFT-LoRA**. Analyzed performance on various test sets, including augmented data, and compared models to identify weaknesses.
- Fine-tuned vits model for **Bengali TTS (উচ্চারণ)**
- Worked on model compression using **Olive**, **Quantization** and **ONNX** for faster loading, inference times and also for offline services
- Worked on **RAG** based chatbot using **GraphRag**, **LightRag**, **Multi-Modal Rag**, **Agentic Rag** and **LangChain** for enhanced AI chatbot ([REVE CHAT](#)) responses.
- Performed sentiment analysis and topic modeling using **BERT** on streaming STT data
- Developed an election commission chatbot using **BERT** and **Bi-LSTM** with better accuracy
- Implemented face liveness detection for NTMC biometric authentication project using **OpenCV**, **CNN** and **MediaPipe** for anti-spoofing
- Developed **Time Series Models** and forecasting pipeline for production quantity and sales amount forecasting.
- **Pytorch | Tensorflow | OpenCV | Langchain | LLM | RAG | Time Series**

Codelab FZC 📅 Jun 2023 – Jan 2024
Software Engineer

- Developed a 2D object detection game and a typing game, where players type sentences within a time limit, using **Unity** for **Android** and **iOS**.
- Contributed to developing an admin panel for a project, focusing on implementing **CRUD** operations in **Laravel**
- **Unity | C# | Lavavel**

RESEARCH EXPERIENCE

Development of a Large Corpus of Spontaneous Speech for Improved Speech Recognition in Bangladeshi Bangla:

We created a 200-hour Bangladeshi Bangla spontaneous speech corpus and fine-tuned the multilingual wav2vec 2.0 model, comparing it with a reading corpus (SUBAK.KO) to enhance ASR performance.

BanSpeech: A Multi-Domain Bangla Speech Recognition Benchmark Toward Robust Performance in Challenging Conditions:

Assessed a Bangla ASR model on BanSpeech, a 7.2-hour benchmark; it struggled with spontaneous speech and high out-of-vocabulary rates, highlighting the importance of domain selection.

- **Paper Link:** ieeexplore.ieee.org/document/10453554

PROJECTS

SUST Student Management System 
Technologies: **Django | Python | HTML | CSS | Bootstrap | MYSQL**
Managed student, faculty data, departments, courses, and faculty assignments; faculty recorded student marks.

Automatic Attendance System Using Face Recognition 
Technologies: **Python | OpenCV | Streamlit**
We used face recognition via a Streamlit web app to mark attendance, collecting five pictures per classmate for identification.

Shoesly an E-commerce Website 
Technologies: **Node.js | React.js | MongoDB | Express.js**
In our app, users, suppliers, admins, and banks have distinct routes for filtering products, confirming orders, and managing transactions between suppliers and customers.

EDUCATION

Shahjalal University of Science & Technology

BSc in Computer Science & Engineering (Graduated)
📅 Feb 2018 – Mar 2023

CGPA: 3.35/4.00

SKILLS

- **Languages:** Python, Java, C, C++, PHP, C#
- **Machine Learning Technologies:** Pytorch, Tensorflow, OpenCV, Langchain, LLM, Scikit-learn, MLflow, CI/CD,
- **Web Technologies:** Django, JavaScript (Node.js, React.js, Vue.js), HTML/CSS, Laravel
- **Database:** MySQL, MongoDB, Firebase
- **Miscellaneous:** Linux, Shell (Bash), LaTeX, Microsoft Office, Docker, Git
- **Problem Solving:** Solved 600+ problems in different online judge including Codeforces ([partha361](#)), Leetcode ([ppartha](#)), lightoj ([parthaghoshu](#)), Vjudge ([2017331039](#)), Stopstalk ([ppaartha](#))

ACHIEVEMENTS

1st Runner up in AI for Bangla 2.0 2023

EXTRA-CURRICULAR ACTIVITIES

CSE Society, SUST
General Secretary, Assistant General Secretary, Executive Member
📅 2019 – 2023

SUST ACM Student Chapter
Joint Secretary
📅 2021 – 2023

REFERENCE

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